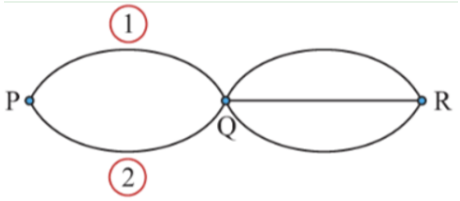
I- Product Principle

Suppose there are three towns P, Q, and R linked by roads as shown. How many different ways are there to get from P to R via Q ?



If we take road 1 from P to Q, then there are 3 possible roads to get from Q to R.

Likewise, if we take road 2 from P to Q, there are the same 3 choices to get from Q to R.

So, there are

$$3 + 3 = 2 \times 3$$

ways of getting from P to R via Q.

Using the same argument for the road connections alongside,



there are

$$3 \times 4 \times 2 = 24$$

ways to get from P to S via Q and R.

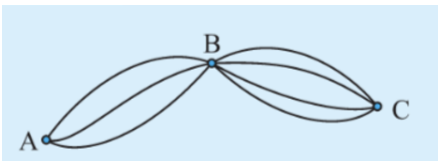
Definition

The **product principle** is :

If there are m different ways of performing an operation, and for each of these there are n different ways of performing a second *independent* operation, then there are mn different ways of performing the two operations in succession.

Example

Three towns A, B, and C are connected by the road network shown. How many different ways are there to get from A to C via B ?

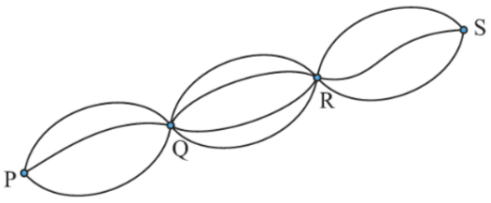


EXERCISE 1

The diagram shows the possible pathways for a bus service between four towns P, Q, R, and S.
How many different routes are possible from:

a Q to S via R

b S to P via R and Q?



EXERCISE 2

Look at the menu below.
How many different combinations are there for a person to order an entrée, a main course, and a dessert?



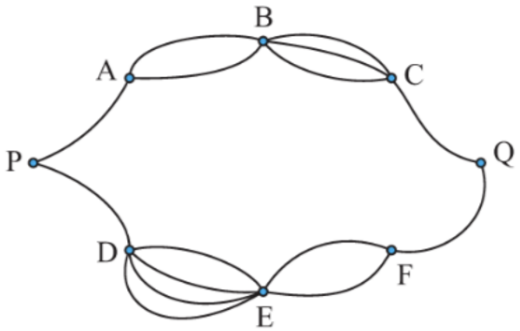
EXERCISE 3

The code for this lock consists of five letters of the alphabet.
How many different codes are possible?



II- Sum Principle

Consider the road system from P to Q.



From A to C there are $2 \times 3 = 6$ pathways. From D to F there are $4 \times 2 = 8$ pathways. So, in total there are $6 + 8 = 14$ different pathways from P to Q.

We added the pathways because we had the choice of going from P to either A *or* D. This is the **sum principle**.

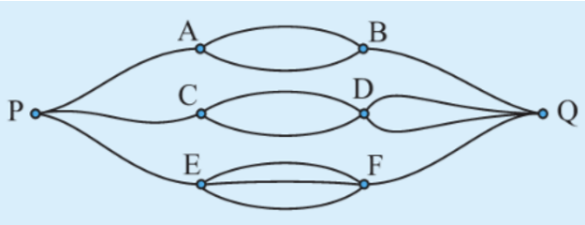
Property

he word **and** suggests **multiplying** the possibilities.

The word **or** suggests **adding** the possibilities.

Example

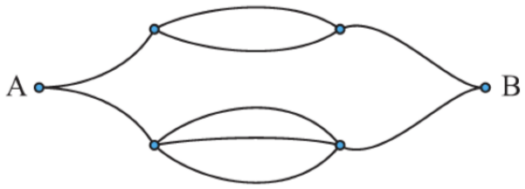
How many different paths lead from P to Q ?



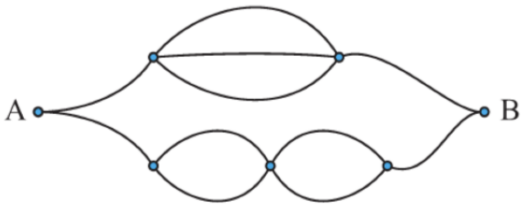
EXERCISE 4

How many different paths lead from A to B?

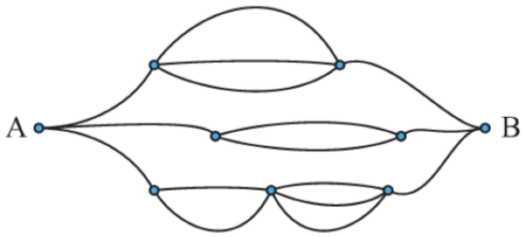
a



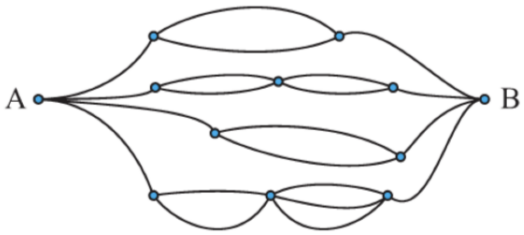
b



c



d

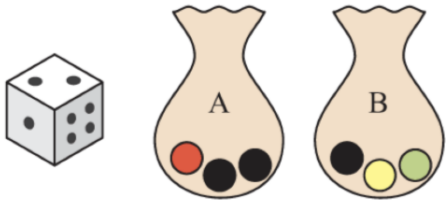


EXERCISE 5

Michelle rolls a die. If the result is even, she picks a ball from bag A. Otherwise, she picks a ball from bag B.
How many ways lead to:

a a yellow ball being picked

b a black ball being picked?



III- Order and arrangements

a) Factorial

Definition

For $n \geqslant 1$, $n!$ is the product of the first n positive integers.

$$n! = n(n-1)(n-2)\ldots 3 \times 2 \times 1$$

$n!$ is read "*n factorial*".

Remark. We define :

$$0! = 1$$

Example

$$11 \times 10 \times 9 \times 8 = \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} = \frac{11!}{7!}.$$

Remark. An alternative definition of factorial numbers is that :

$$n! = n \times (n-1)! \quad \text{for } n \geqslant 1.$$

Example

1. $9 \times 8 \times 7$
2. $\dfrac{11 \times 10 \times 9 \times 8}{4 \times 3 \times 2 \times 1}$

EXERCISE 6

Express in factorial form:

a $4 \times 3 \times 2 \times 1$ b $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$ c 6×5

d $8 \times 7 \times 6$ e $10 \times 9 \times 8 \times 7$ f $15 \times 14 \times 13 \times 12$

g $\dfrac{9 \times 8 \times 7}{3 \times 2 \times 1}$ h $\dfrac{13 \times 12 \times 11 \times 10}{4 \times 3 \times 2 \times 1}$ i $\dfrac{15 \times 14 \times 13 \times 12 \times 11}{5 \times 4 \times 3 \times 2 \times 1}$

EXERCISE 7

Simplify without using a calculator:

a $\dfrac{7!}{6!}$ b $\dfrac{8!}{6!}$ c $\dfrac{12!}{10!}$ d $\dfrac{120!}{119!}$ e $\dfrac{10!}{8! \times 2!}$ f $\dfrac{100!}{98! \times 2!}$

EXERCISE 8

Simplify:

a $\dfrac{n!}{(n-1)!}$ b $\dfrac{(n+2)!}{n!}$ c $\dfrac{(n+1)!}{(n-1)!}$

EXERCISE 9

Write as a product by factorising:

a $5! + 4!$ b $11! - 10!$ c $5! + 7!$ d $12! - 10!$

e $9! + 8! + 7!$ f $7! - 6! + 8!$ g $12! - 2 \times 11!$ h $3 \times 9! + 5 \times 8!$

EXERCISE 10

Simplify by factorising:

a $\dfrac{12! - 11!}{11}$ b $\dfrac{10! + 9!}{11}$ c $\dfrac{10! - 8!}{89}$ d $\dfrac{10! - 9!}{9!}$

e $\dfrac{6! + 5! - 4!}{4!}$ f $\dfrac{n! + (n-1)!}{(n-1)!}$ g $\dfrac{n! - (n-1)!}{n-1}$ h $\dfrac{(n+2)! + (n+1)!}{n+3}$

b) Permutations

Definition

A permutation of a group of symbols is *any arrangement* of those symbols in a *definite order*.

Example

Consider the three symbols P, Q, and R. A permutation which uses all of these symbols is RQP.

We say they have been taken "three at a time".

There are in fact six different permutations on the symbols P, Q, and R taken three at a time.

They are :

$$PQR, \quad PRQ, \quad QPR, \quad QRP, \quad RPQ, \quad RQP.$$

EXERCISE 11 List the permutations on the symbols X, Y, and Z when they are taken :

1. one at a time
2. two at a time
3. three at a time

Property

he number of permutations on n different symbols taken r at a time is

$$\frac{n!}{(n-r)!}.$$

EXERCISE 12

Find the number of permutations there are on the symbols:

a A, B, C, D, E, and F taken 3 at a time

b P, Q, R, S, T, U, and V taken 5 at a time

c A to Z taken 15 at a time.

EXERCISE 13

A football competition is organised between 8 teams. In how many ways is it possible to fill the top four places on the competition ladder?

Example

The alphabet blocks A, B, C, D, and E are placed in a row in front of you.

1. How many different permutations are there ?
2. How many permutations end in C ?
3. How many permutations have the form $\ldots A\ B \ldots$?
4. How many permutations begin and end with a vowel (A or E) ?

EXERCISE 14

A 4-digit number is made up using the digits 1, 2, 3, 4, 5, and 6 at most once each. How many different numbers can be formed if:

a there are no other restrictions

b the number must be divisible by 5

c the number must start with 6 and end with 2?

EXERCISE 15

13 Sprinters A, B, C, and D are members of a 4×400 m relay team. In how many ways can the sprinters be ordered if:

a there are no restrictions

b B must sprint last

c D must not sprint last

d C must sprint either first or second

e A must be followed directly by B?

Example

There are 6 different books arranged in a row on a shelf. In how many ways can two of the books, A and B, be together ?

EXERCISE 16

Consider the letters of the word MONDAY. How many different permutations of four letters can be chosen if:

a there are no restrictions

b at least one vowel (A or O) must be used

c the two vowels are not together?

EXERCISE 17

Nine cupcakes are each iced with a different whole number from 1 to 9. Five children are allowed to take one cupcake each. In how many different ways can this be done if:

a there are no restrictions

b the first three children all decide that they will choose an even numbered cupcake?

EXERCISE 18

Alice has booked ten adjacent front-row seats for a basketball game for herself and nine friends.

a How many different arrangements are possible if there are no restrictions?

b Due to a severe snowstorm, only five of Alice's friends are able to join her for the game. In how many different ways can they be seated in the 10 seats if:

i there are no restrictions

ii they all sit together with no empty seats between

iii any two of Alice's friends sit either side of her?

c) Permutations and Repetition

Example

Consider the different orderings of the symbols A, A, A, B, and C.

If the 5 symbols were all different, they could be ordered in $5!$ ways.

Consider the permutation $ABAAC$. If we label the 3 As with subscripts, we see there are 6 possible orderings of the As which result in this permutation :

$$A_1BA_2A_3C \quad A_2BA_1A_3C \quad A_3BA_1A_2C$$
$$A_1BA_3A_2C \quad A_2BA_3A_1C \quad A_3BA_2A_1C$$

This is true for all of the possible permutations, so the total number of distinguishable permutations is :

$$\frac{5!}{3!} = 20.$$

Property

uppose there are n_1 of the first symbol, n_2 of the second symbol, ..., and n_r of the rth (and final) symbol.

The total number of distinguishable permutations is :

$$\frac{(n_1 + n_2 + n_3 + \ldots + n_r)!}{n_1!\, n_2!\, n_3! \ldots n_r!}.$$

Example

Consider the word **CALCULUS**. How many permutations can be formed :

1. using all of the letters,
2. which start with A and end with S,
3. which start with C and end with L ?

EXERCISE 19

Two identical Chemistry books and three identical Mathematics books are placed in a pile. In how many distinguishable ways can they be stacked if:

a there are no restrictions

b the books in each subject are kept together?

EXERCISE 20

One red, two white, and three blue marbles are placed in a line. In how many distinguishable ways can this be done if:

a white marbles are at either end b blue marbles are at either end

c the red marble is at one end and a white marble is at the other?

EXERCISE 21

A Mahjong set contains:

- 4 of each of 9 tiles, in each of 3 suits
- 4 of each of 3 dragons
- 4 of each of 4 winds
- 1 of each of 4 birds, in each of 2 colours.

Explain why the total number of orders in which the tiles can be selected is $\frac{144!}{(4!)^{34}}$.

d) Combinations

Definition

A combination is a selection of objects without regard to order or arrangement.

Example

The possible teams of 3 people selected from A, B, C, D, and E are :

$$ABC \quad ABD \quad ABE \quad ACD$$
$$ACE \quad ADE \quad BCD \quad BCE$$
$$BDE \quad CDE$$

We only list ABC and not ACB, BAC, BCA, CAB, and CBA.

Remark. The number of *permutations* on 5 letters taken 3 at a time is $5 \times 4 \times 3 = 60$, because each combination has $3 \times 2 \times 1$ possible permutations.

Remark. ${}^5C_3$ is read "5 choose 3." It can also be written as C_3^5 or $\binom{5}{3}$.

Property

he number of combinations on n distinct symbols taken r at a time is :

$${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

EXERCISE 22

Evaluate:

a $\binom{3}{2}$ b $\binom{6}{4}$ c $\binom{5}{1}$ d $\binom{7}{2}$

Use your calculator to check your answers.

EXERCISE 23

a List the different teams of 2 that can be chosen from a squad of 5 players named P, Q, R, S, and T.

b Check that $\binom{5}{2}$ gives the total number of teams.

EXERCISE 24

a Evaluate:

i $\binom{5}{0}$ ii $\binom{5}{5}$ iii $\binom{9}{0}$ iv $\binom{9}{9}$

b Use combinations to explain why, for any $n \in \mathbb{N}$:

i $\binom{n}{0} = 1$ ii $\binom{n}{n} = 1$

EXERCISE 25

a Evaluate:

i $\binom{7}{2}$ and $\binom{7}{5}$ ii $\binom{10}{3}$ and $\binom{10}{7}$

b Use the formula $\binom{n}{r} = \dfrac{n!}{r!(n-r)!}$ to show that $\binom{n}{k} = \binom{n}{n-k}$ for any $n, k \in \mathbb{N},\ 0 \leqslant k \leqslant n$.

c Use combinations to explain why this is true.

EXERCISE 26

Find k such that $\binom{9}{k} = 4\binom{7}{k-1}$.

EXERCISE 27

Write as a polynomial in n, including an appropriate doma

a $\binom{n}{n-2}$ b $\binom{n}{4}$

EXERCISE 28

A physics exam contains 7 questions. Students must answer both questions 1 and 2, and any 3 of the remaining questions. How many different selections are possible?

Example

An equestrian team of 4 is to be chosen from 7 male and 6 female riders. How many different teams can be chosen if :

1. there are no restrictions,
2. there must be two riders of each sex,
3. at least one rider of each sex is required ?

EXERCISE 29

A committee of 4 singers is chosen from 7 tenors and 9 sopranos.

Find the number of ways of selecting the committee if:

a there are no restrictions

b it must contain 2 tenors and 2 sopranos

c it must contain at least 2 sopranos.

EXERCISE 30

A soccer team consists of 11 players. The team must be sorted into 3 attackers, 3 midfielders, 4 defenders, and 1 goalkeeper for today's match.

a How many possible ways are there to do this?

b Melissa is chosen to play goalkeeper. Linda wants to play attacker, and Robyn does not want to play defender. How many combinations are now possible?

EXERCISE 31

How many 4-digit numbers can be constructed in which the digits are in ascending order from left to right? You cannot start a number with 0.

EXERCISE 32

10 points are located on [PQ], 9 on [QR], and 8 on [RP].

All possible lines connecting these 27 points are drawn. Determine the maximum possible number of points of intersection of these lines which lie within triangle PQR.

